AC LIGHTS CONTROLLING USING IR REMOTE

INTRODUCTION :

In modern households and workplaces, automation is becoming a key aspect of convenience and efficiency. This project demonstrates a simple and practical home automation system to control three AC lights using an IR (Infrared) remote and an Arduino. The system leverages the Arduino microcontroller to decode infrared signals from a remote control and activate or deactivate the connected lights via a relay module. This project not only highlights the use of microcontrollers for home automation but also emphasizes the integration of low-power digital electronics with high-power electrical appliances. By eliminating the need for physical switches, the system provides a user-friendly interface for operating devices wirelessly, making it ideal for applications in smart homes and classrooms.

KEY OBJECTIVES

1. To wirelessly control multiple AC lights using an IR remote.
2. To demonstrate the use of relays for isolating low-voltage control circuits from high-voltage AC loads.
3. To create a scalable system that can be extended to control additional devices or appliances.

PROJECT FEATURES

1. Wireless Operation: Lights are controlled using an IR remote, allowing users to operate them from a distance.
2. Energy Efficient: The system ensures that lights are turned ON or OFF as needed, reducing energy wastage.
3. Cost-Effective: Utilizes readily available and affordable components like Arduino, IR modules, and relays.
4. Compact and Safe Design: Ensures electrical safety by isolating high-voltage AC circuits from the low-voltage control system.

APPLICATIONS

1. Home Automation: Enables remote control of lights, fans, or other household appliances.
2. Classrooms and Offices: Facilitates easy control of lighting systems without manual switches.

3. Accessibility: Assists individuals with mobility issues by providing remote operation of electrical devices.

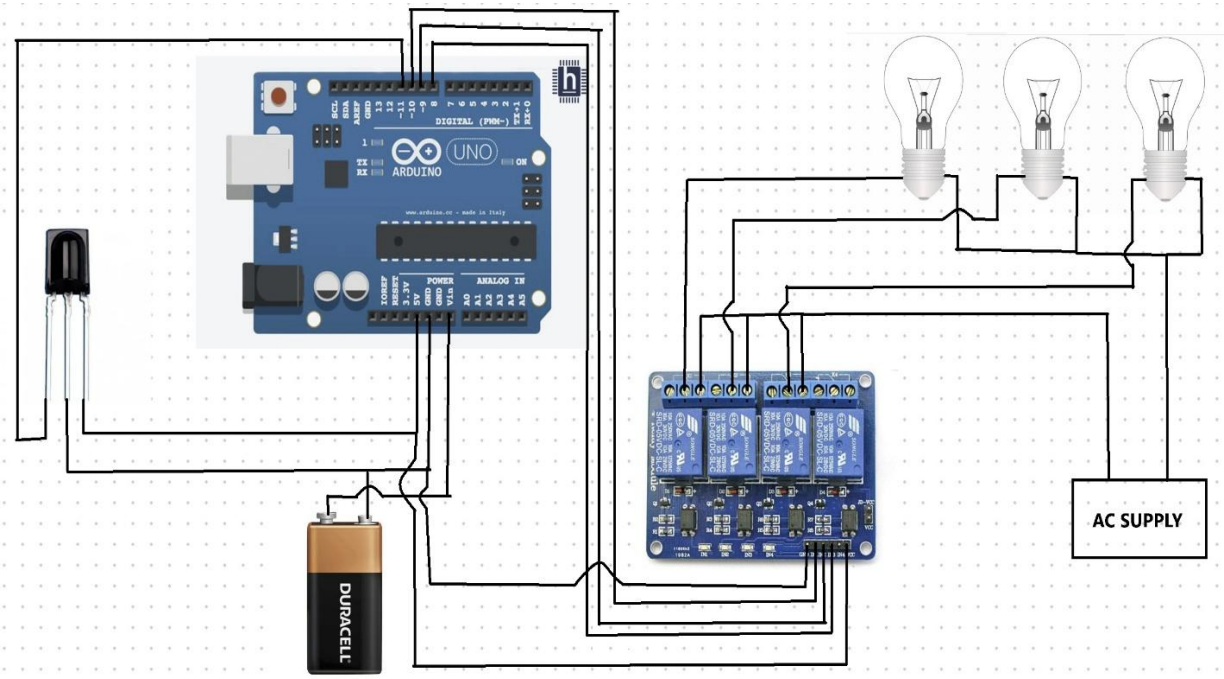
This project is an excellent starting point for understanding the basics of home automation and offers opportunities for further enhancements, such as integrating Wi-Fi or Bluetooth for smart home capabilities.

APPARATUS :

1. Arduino Board (Uno)
2. IR Receiver Module (TSOP1838)
3. Relay Module (4-channel relay)
4. AC Lights
5. Power Supply:
   Arduino: USB or 9V DC supply.
   AC lights: 230V AC
6. IR Remote
7. Connecting Wires
8. Breabboard

BLOCK DIAGRAM :

CIRCUIT DIAGRAM :



IR REMOTE DECODING:

IR remote decoding involves capturing and interpreting signals from an infrared remote control using a TSOP1838 IR receiver and an Arduino. By using the IR remote library, the Arduino can read and display the unique hex codes corresponding to each button press. These codes can then be used to control various devices, such as a DC motor speed controller. This process allows for wireless control, making it ideal for automation projects.

CODE :

```
#include <IRremote.h>

const int RECV_PIN = 11;
IRrecv irrecv(RECV_PIN);
decode_results results;
```

```
void setup() {
  Serial.begin(9600);
  irrecv.enableIRIn();
  Serial.println("TSOP1838 IR Receiver Ready");
}

void loop() {
  if (irrecv.decode(&results)) {
    Serial.print("IR Code: ");
    Serial.println(results.value, HEX);
    irrecv.resume();
  }
}
```

sketch_apr3a.ino

```
#include <IRremote.h>

const int RECV_PIN = 11; // TSOP1838 OUT pin connected to D2
IRrecv irrecv(RECV_PIN);
decode_results results;

void setup() {
    Serial.begin(9600);  // Start Serial Monitor
    irrecv.enableIRIn(); // Start the IR receiver
    Serial.println("TSOP1838 IR Receiver Ready");
}

void loop() {
    if (irrecv.decode(&results)) {
        Serial.print("IR Code: ");
        Serial.println(results.value, HEX); // Print the received IR code in HEX format
        irrecv.resume(); // Receive the next value
    }
}
```

Output Serial Monitor ×

Message (Enter to send message to 'Arduino Uno' on 'COM7')

```
IR Code: 128D5751
IR Code: 1FE40BF
IR Code: 1FEC03F
IR Code: FFFFFFFF
IR Code: 1FE50AF
IR Code: 1FED827
IR Code: 1FEF807
IR Code: 1FE30CF
IR Code: 1FEB04F
IR Code: 1FE708F
```

BUTTON HEX CODES:

OPERATION	HEX CODE	BUTTON
ON/OFF BULB 1	0x1FE50AF	Button "1"
ON/OFF BULB 2	0x1FED827	Button "2"
ON/OFF BULB 3	0x1FEF807	Button "3"
OFF ALL BULBS	0x1FEE01F	Button "0"
ON ALL BULBS	0x1FE48B7	Button "⏻"

METHODOLOGY:

Step 1: Gather components

Step 2: Circuit Connections

1. Connect the IR Receiver to Arduino
   - VCC → Arduino 5V
   - GND → Arduino GND
   - OUT → Arduino D11 (or any other digital pin)
2. Connect the Relay Module
   - Relay Inputs:
     - VCC → Arduino 5V
     - GND → Arduino GND
     - IN1 → Arduino D8 (Control for Light 1)
     - IN2 → Arduino D9 (Control for Light 2)
     - IN3 → Arduino D10 (Control for Light 3)
   - Relay Outputs (AC Side):

- COM Terminal → Connect to the live wire from the AC mains.
- NO Terminal → Connect to the live input of each light:
  - NO1 → Light 1 live input
  - NO2 → Light 2 live input
  - NO3 → Light 3 live input
- Connect the neutral wire directly to all lights.

3. Connect the Arduino Power Supply

- Use either a USB cable or a DC adapter (5V or 9V) to power the Arduino.

Step 3: Upload the Code

Step 4: Test the IR Remote

1. Open the Serial Monitor in the Arduino IDE.
2. Point the IR remote at the IR receiver and press buttons to see the **hexadecimal codes** printed.
3. Replace the codes in the case statements with the actual codes for your remote.

Step 5: Assemble and Test the Circuit

1. **Check Connections:**
   - Ensure all AC connections are insulated.
   - Verify the relay module connections to the lights.
2. **Power the System:**
   - Connect the Arduino to the power supply.
   - Plug the AC lights into their respective sockets.
3. **Control the Lights:**
   - Press buttons on the IR remote to toggle the lights ON or OFF.

CODE:

```
// Include the IRremote library
#include <IRremote.h>

// Pin Definitions
#define RECV_PIN 11 // Pin connected to IR Receiver OUT
#define RELAY2 10    // Pin connected to Relay 2
#define RELAY3 9    // Pin connected to Relay 3
#define RELAY4 8   // Pin connected to Relay 4

// Create an IR receiver object
IRrecv irrecv(RECV_PIN);
decode_results results; // Stores received IR signal

void setup() {
  // Start Serial Monitor for debugging
  Serial.begin(9600);

  // Initialize IR receiver
  irrecv.enableIRIn();

  // Set relay pins as OUTPUT
  pinMode(RELAY2, OUTPUT);
  pinMode(RELAY3, OUTPUT);
  pinMode(RELAY4, OUTPUT);

  // Turn off all relays initially (active LOW relays)
  digitalWrite(RELAY2, HIGH); // Relay OFF
  digitalWrite(RELAY3, HIGH); // Relay OFF
  digitalWrite(RELAY4, HIGH); // Relay OFF
}

void loop() {
  // Check if an IR signal is received
  if (irrecv.decode(&results)) {
    // Print the received code to Serial Monitor in HEX
format
    Serial.println(results.value, HEX);

    // Compare received code and control relays
    switch (results.value) {
```

```
      case 0x1FE50AF: // Button for Light 1
        digitalWrite(RELAY2, !digitalRead(RELAY2));
// Toggle Light 1
        break;
      case 0x1FED827: // Button for Light 2
        digitalWrite(RELAY3, !digitalRead(RELAY3));
// Toggle Light 2
        break;
      case 0x1FEF807: // Button for Light 3
        digitalWrite(RELAY4, !digitalRead(RELAY4));
// Toggle Light 3
        break;
      case 0x1FEE01F: // Turn OFF all lights
        digitalWrite(RELAY2, HIGH);
        digitalWrite(RELAY3, HIGH);
        digitalWrite(RELAY4, HIGH);
        break;
      case 0x1FE48B7: // Turn ON all lights
        digitalWrite(RELAY2, LOW);
        digitalWrite(RELAY3, LOW);
        digitalWrite(RELAY4, LOW);
        break;
    }

    // Resume IR receiver for the next signal
    irrecv.resume();
  }
}
```

CONCLUSION:

The project "Controlling AC Lights with an IR Remote Using Arduino" provides a practical and cost-effective solution for basic home automation. By integrating an IR receiver, Arduino microcontroller, and a relay module, the system allows for wireless control of AC lights, enhancing convenience and accessibility for users. The design ensures safety by isolating the low-voltage control circuit from the high-voltage AC load through relays, and it operates reliably with quick response to the IR remote commands.

Although the system is functional and easy to implement, it has limitations, including the short range of the IR remote, which requires line-of-sight communication. Additionally, the system lacks advanced features such as

automation, scheduling, or voice control. The current design can also only control a limited number of lights due to the relay module's channel capacity.

Future improvements could include extending the control range by incorporating RF or Wi-Fi, as well as adding automation through sensors or voice control. These enhancements would make the system more versatile and in line with modern smart home technologies.

Overall, this project serves as a solid foundation for home automation, offering valuable insights into how simple electronic components can be used for practical applications. It highlights the potential for further development and innovation in smart home solutions.